PROJECT PRESENTATION

Convolutional Autoencoder for Image Colorization

Ariana Sauvan

UNIVERSITY OF CA' FOSCARI

Image & Video Understanding • June 2026

MOTIVATION

Why Automatic Colorization?

Manual Workload

Historically, colorization has been a painstaking, manual, and slow artistic process requiring deep historical research and expert photo editing skills.

Convolutional Autoencoder

It is a simple baseline that helps understand the fundamentals before moving to more complex models

Self-Supervised

By splitting standard color images into grayscale and color components, models can train without labels.

Grayscale to Color Task

Color Space Mapping

The core objective is to translate a single-channel grayscale representation into multi-channel color coordinates. Standard RGB is complex, making direct regression difficult.

- **Input:** L (Lightness) channel representing brightness.
- **Target:** AB channels containing chrominance data.

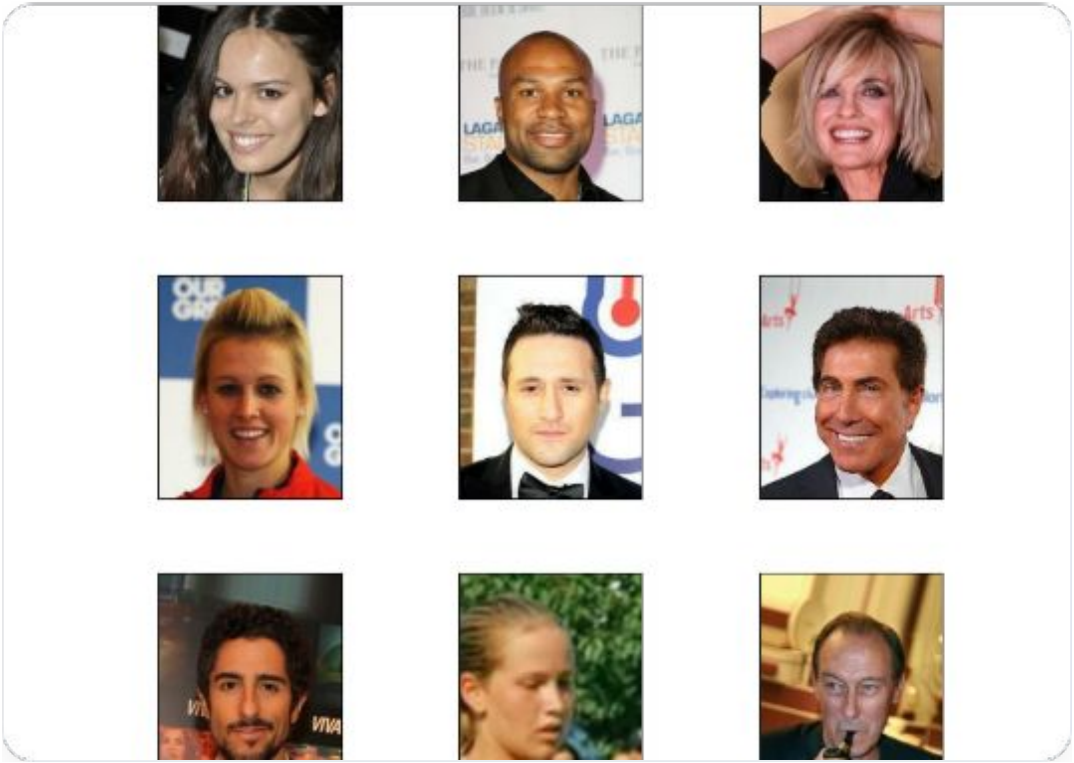
Key Challenges

Colorization is an inherently ambiguous problem. A single grayscale object can have multiple equally valid and realistic color options. The model must guess the most plausible one without explicit color labels.

- **Averaging Effect:** Standard pixel-wise losses (like MSE) encourage the model to average all possible colors.
- **Ambiguous Nature:** Many objects have multiple plausible color modes, the model struggles to commit to one.

The CelebA Face Dataset

- **Diverse Portrait Gallery:** Over 200k high-resolution facial images containing significant diversity in pose, hair color, age, and backgrounds.
- **Structured Alignment:** Centered and aligned face crops make the learning task highly reliable for establishing structural color priors.
- **Clean Evaluation Splits:**
  - Train Split: 162,770 samples
  - Validation Split: 19,867 samples
  - Test Split: 19,962 samples



PREPROCESSING PIPELINE

Data Preprocessing

1. Image Resizing

All inputs are scaled down to 64 × 64 pixels to keeps the model lightweight and fast to train while being large enough to keep the facial structure.

2. LAB Color Space

RGB mixes brightness and color in all three channels, which makes the learning task harder. LAB separates them cleanly.

3. Range Normalization

L values are normalized into the interval [0, 1] for inputs. Color-carrying AB values are normalized into [-1, 1] .

Model Architecture

Convolutional Autoencoder (CAE)

- Encoder: 4 convolutional blocks (stride-2 downsampling)
- Bottleneck: compressed latent representation
- Decoder: 4 transposed-conv blocks (upsampling)
- Output: AB channels (Tanh activation)
- No skip connections (baseline architecture)

Encoder Specifications

- **Four Downsampling Blocks:** Progressive reduction of spatial resolution using Conv2d layers with stride 2, allowing the encoder to compress the input while increasing feature depth.
- **Symmetric Block Units:** Each block integrates a Conv2d (3×3, stride 2) layer, followed directly by Batch Normalization and a stabilizing ReLU activation.
- **Spatial Latent Progression:** Reduces input spatial resolution sequentially from 64 → 32 → 16 → 8, producing a compact latent representation.

Latent Bottleneck

The resulting representation at the narrowest section of the model is a high-dimensional feature map of size:

512 × 4 × 4

This bottleneck forces the model to discard fine textures and instead encode semantic facial features such as skin tone, hair structure, and global face layout.

Decoder Specifications

Spatial Reconstruction

The decoder expands the 4×4 latent representation back to 64×64 through successive transposed convolutions, gradually restoring spatial structure.The final output shape is :

2 × 64 × 64

The decoder outputs 2 channels (a and b), representing the red–green and blue–yellow chrominance components of the LAB color space.

- **Four Upsampling Blocks:** Reconstructs structural information symmetrically using ConvTranspose2d (4×4 kernel, stride of 2) mirroring the encoder.
- **Block Structure :** Intermediate blocks include: ConvTranspose2d to perform upsampling, BatchNorm to stabilize activations and a ReLU function to maintain smooth gradient flow
- **Tanh Boundary Output:** The final block uses a Tanh activation function to cleanly constrain predictions into the target range [-1, 1].

Training

Hyperparameter / Metric	Configuration Detail	Purpose
Optimizer	Adam (Learning Rate = 0.001)	Robust and works well for image tasks.
Loss Function	Mean Squared Error (MSE)	Standard loss for colorization baselines.
Epoch Count / Batch Size	20 Epochs / 64 Batch Size	Enough for convergence on 64×64 images.
Hardware Environment	Google Colab T4 GPU	Accelerates parallelized tensor matrix calculations.
Training Duration	~2 hours 15m	Consistent per-epoch runtime (~6–7 minutes) enabling reliable convergence over 20 epochs.

MODEL RESULTS

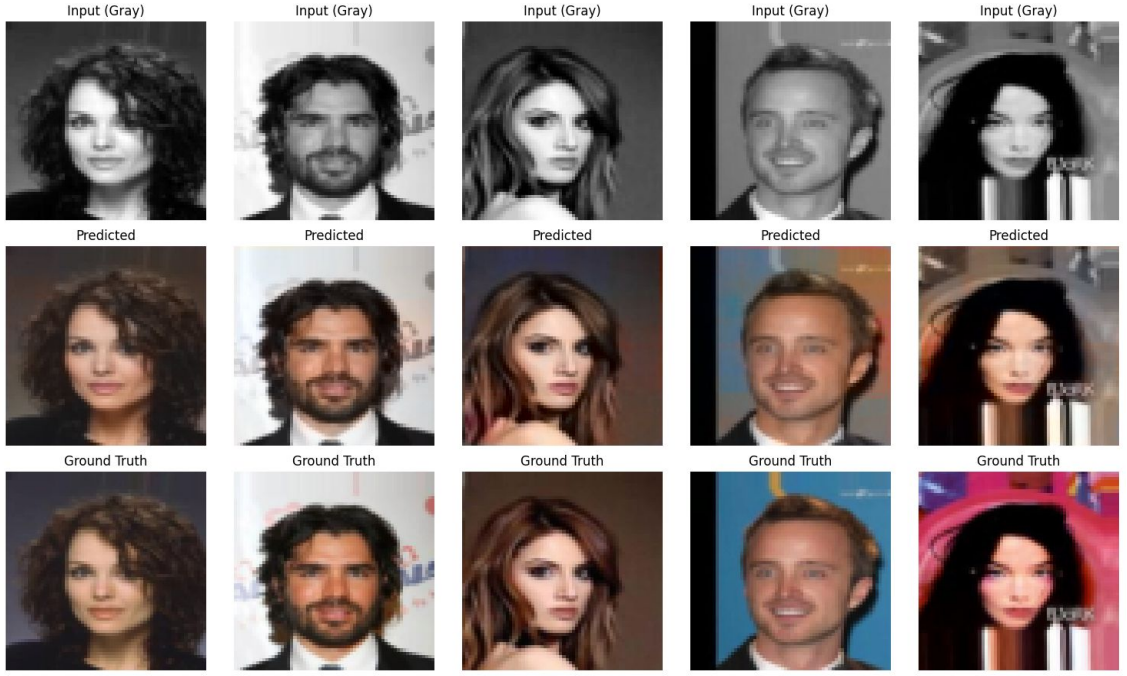
Qualitative Evaluation

Key Strengths

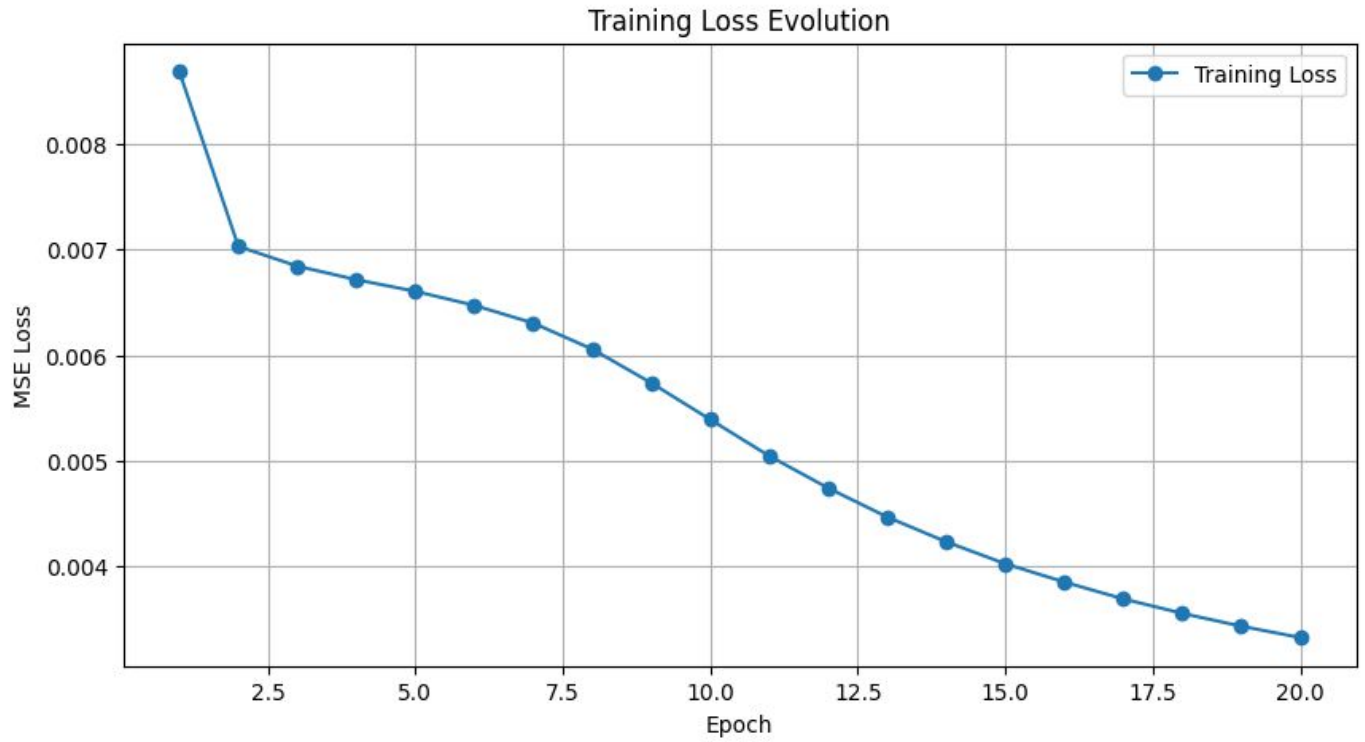
Consistent facial colorization — The model predicts skin tones, hair colors, and other facial regions.
Coherent outputs — Reconstructed images show stable color transitions due to the CAE's bottleneck representation.
Semantic feature capture — The encoder learns high-level cues, enabling plausible color assignments even without texture details.

Limitations

Desaturated backgrounds — Regions with high ambiguity tend to appear grayish because MSE encourages average color predictions.
Loss of fine detail — The 4×4 bottleneck removes high-frequency textures, leading to blurry areas.
Limited generalization — Trained only on faces, the model struggles with objects or scenes outside the CelebA domain.



Quantitative Analysis



Training Loss Curve

Smooth and consistent decrease across 20 epochs, no sudden spikes and leads to a stable optimization.

Final loss: 0.0033

Validation loss: 0.0080

Gap with training loss: 0.0046

EVALUATION

Limitations & Future Work

Limitations

- **Standard MSE:** Leads to desaturated, averaged color choices in ambiguous regions.
- **Missing Connections:** The lack of skip connections reduces image sharpness and high-frequency details.
- **Loss of fine details:** The 4×4 bottleneck removes high-frequency textures, causing oversmoothing.
- **Dataset Bias:** Trained on CelebA, causing performance drops on non-face domains.

Future Enhancements

- **U-Net Layout:** Implement structural skip connections between encoder and decoder.
- **Perceptual Loss:** Move from pixel-level MSE to perceptual losses.
- **GAN Architecture:** Employ adversarial training (cGAN) for realistic texture modes.
- **Higher Resolutions:** Scale the inputs to 256 × 256 pixels.
- **Dataset Diversity:** Train on larger, more diverse datasets for better generalization beyond faces.

WRAP UP

Conclusion

- **Semantic Color Learning :** Demonstrates that a simple CAE can successfully learn global semantic color representations on structured datasets.
- **Unsupervised Viability:** Self-supervised colorization offers scaling without expensive external annotation pipelines.
- **Development Baseline:** Provides a robust structural baseline for more advanced conditional architectures like GANs or diffusion frameworks.

Key Takeaway

"The Convolutional Autoencoder serves as an essential stepping stone — highlighting how latent structures store human face priors, while clearly outlining the need for skip connections to avoid blurriness."

Thank you for your attention !

Thank you for your attention!

ariana.sauvan@unive.it Department of Computer Science github.com/Ariana-Sauvan